Draft for discussion

Eastbrook Water Treatment Works Upgrade: Gate 3 request for £194m

Investment decision paper for the Calder Valley Water Investment Committee

Calder Valley Water Investment Committee | 12 November 2026 | Eastbrook Project Team

Confidential and proprietary. For internal discussion only.

We ask the Committee to approve £194m for a new membrane plant, funded at P80 with all Gate 2 conditions closed

The ask

- Approve **£194m** at Gate 3, set at P80
- Includes £18m risk allowance and £12m management reserve, held separately

The case

- The 1974 works cannot meet the 2029 treatment standard
- Serves 214,000 customers; no spare capacity if a stream fails

The preferred option

- New membrane plant on the adjacent site scores **3.2 of 4**
- Rebuilding on the current site scores 2.6 and takes 14 months longer

The risks

- Ground conditions and membrane supply drive most of the uncertainty
- Without the extra £12m, the budget has only a 50% chance of holding

Gate 2 conditions

- Four of four closed: planning consent, land purchase, design freeze, supplier price check

1. P50 and P80 are the costs with a 50% and 80% chance of not being exceeded. Estimate is AACE Class 3 (accuracy -10% to +20%), 2026 Q3 prices, excluding recoverable VAT.

Source: Eastbrook project team, Class 3 estimate and QRA, October 2026. Fictional example.

Eastbrook has passed three gates, and Gate 3 is the decision to release construction funding



Strategic need

Mar 2024
Approved

Options

Feb 2025
Approved, £8m development

Concept design

Jun 2026
Approved with 4 conditions

Investment decision

Nov 2026

- Class 3 estimate at P80
- Preferred option evidenced
- Gate 2 conditions closed
- Funding source confirmed
- Delivery plan with float

Handover

Mar 2029

- Treatment standard met
- Operator trained

◆ Passed ◆ Current gate ◆ Future

1. Gates follow the Calder Valley Water capital investment policy.
Source: Eastbrook project team, Class 3 estimate and QRA, October 2026. Fictional example.

The new membrane plant scores highest at 3.2 of 4, ahead of a rebuild on the current site

Long list (9) → Short list (4) → **Evaluated** → Recommendation

	Weight	**Option A** Do minimum	**Option B** Refurbish existing works	**Option C** New membrane plant, adjacent site	**Option D** Rebuild on current site
Meets 2029 treatment standard	30%	○	◑	●	●
Whole-life cost	25%	●	◑	◕	◔
Deliverability	20%	◕	◕	◕	◑
Resilience	15%	○	◔	◕	◕
Carbon	10%	◔	◑	◑	◕
Weighted score		**1.7**	**2.1**	**3.2**	**2.6**

○ Weak ◔ ◑ ◕ ● Strong

Rationale

- Option A fails the 2029 standard and risks regulator enforcement
- Option C keeps the old works running during construction
- Option D matches C on resilience but needs a 14-month outage plan
  - Temporary supply would cost an extra £21m
- Option B meets only part of the standard

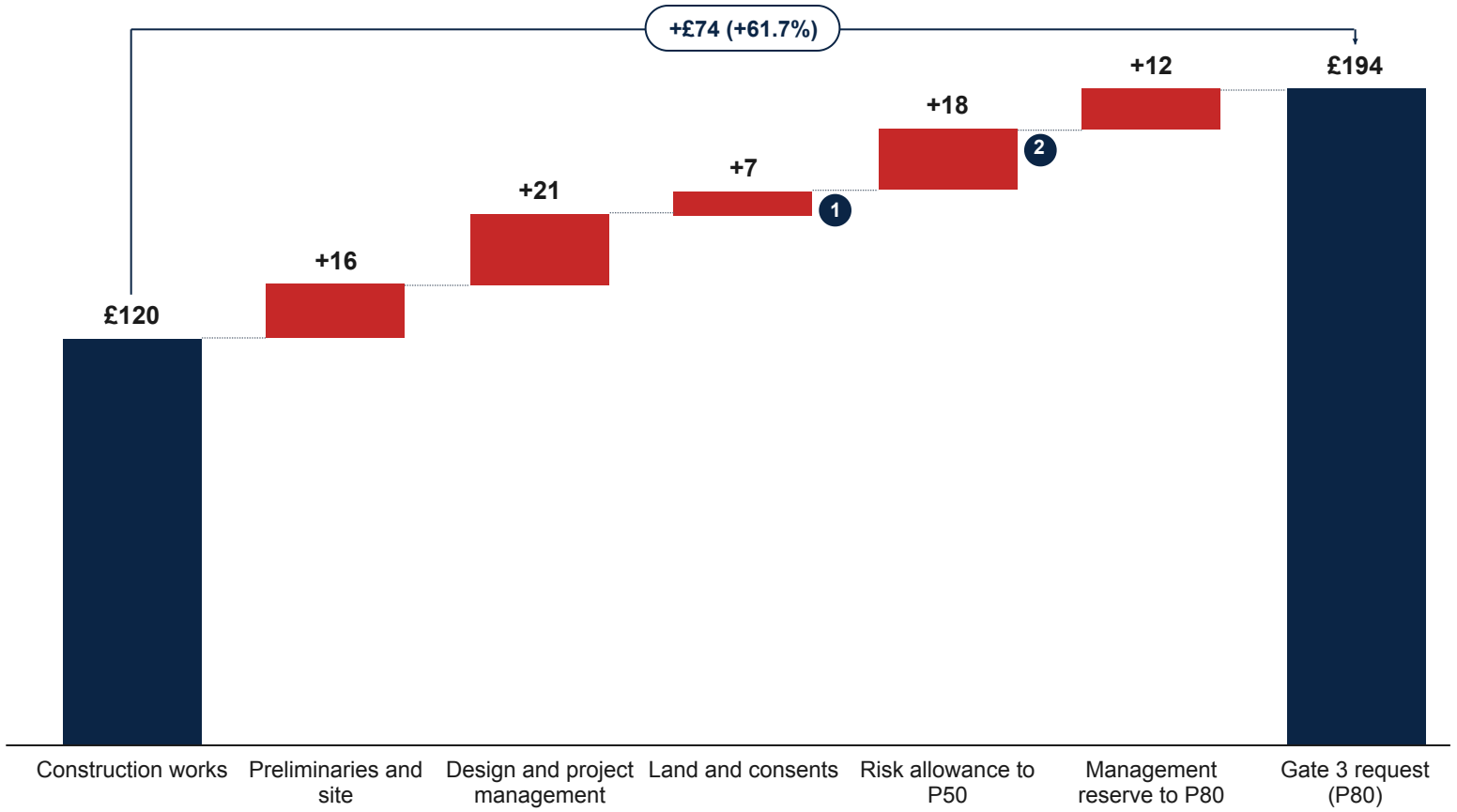
We recommend Option C: build a new membrane plant on the adjacent site

1. Scores 0 (poor) to 4 (strong), set by the project team and challenged by the independent assurance panel on 22 September 2026. Scores are weighted by the percentages shown.
Source: Eastbrook project team, Class 3 estimate and QRA, October 2026. Fictional example.

The £194m request builds up from £120m of works, with risk allowances added separately to reach P80

Cost build-up to the Gate 3 request

£m, 2026 Q3 prices, excl. VAT, AACE Class 3 estimate (-10% / +20%)



What this shows

1. The base estimate is **£164m** before any risk allowance.
2. Risk allowance of **£18m** takes the cost to P50, £182m.
3. The extra **£12m** reserve lifts confidence from 50% to 80%.

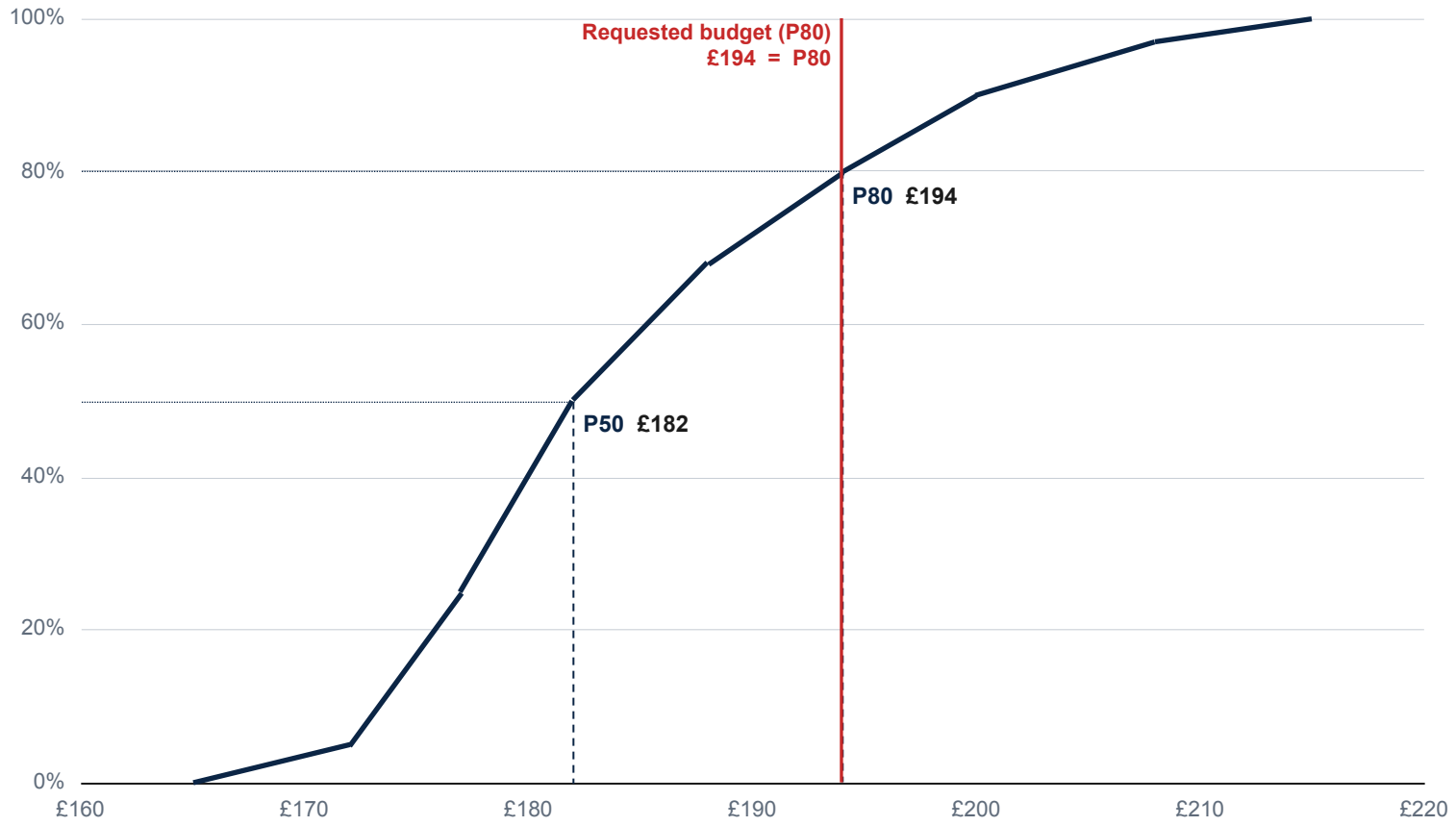
1. P50 = £182m after the £18m risk allowance. Management reserve of £12m takes the total to P80.
2. Contingency and management reserve are held by the Investment Committee, not the project.

Source: Eastbrook project team, Class 3 estimate and QRA, October 2026. Fictional example.

At £194m the budget has an 80% chance of holding, against 50% at the £182m base case

Probability that outturn cost is at or below

£m, 2026 Q3 prices, excl. VAT



Implication for Gate 3

1. Funding at P80 meets the Committee's confidence policy.
2. The largest drivers are ground conditions and membrane supply lead times.

1. Monte Carlo cost risk analysis, 5,000 iterations, risk register version 7 dated 9 October 2026, with correlation between ground and civils risks.
Source: Eastbrook project team, Class 3 estimate and QRA, October 2026. Fictional example.

We ask the Committee for three approvals and one note to protect the P80 funding

#	Decision required	Recommendation and impact	Owner	By
1	**Approve £194m at P80 for Option C, released in two tranches**	Approve. Tranche 1 of £150m now; tranche 2 of £44m at construction contract award	Investment Committee	12 Nov
2	**Authorise award of the construction contract up to £136m**	Approve once the membrane supplier price check is repeated in January 2027	Director of Capital Delivery	29 Jan
3	**Hold the £12m management reserve centrally**	Approve. Release only by Committee decision, reported quarterly	Chief Financial Officer	12 Nov
4	**Note the Gate 4 criteria and the March 2029 handover date**	Note. Progress reported to the Committee each quarter against the P80 budget	Director of Capital Delivery	Feb 27

Source: Eastbrook project team, Class 3 estimate and QRA, October 2026. Fictional example.